

PLAYER SIMILARITY ANALYSIS

Using Unsupervised Machine Learning and Cosine Similarity

Machine Learning Major Project
Case Study 108

Project Information	Details
Project Type	Machine Learning Major Project
Domain	Sports Analytics / Football
Problem Type	Unsupervised Learning
Primary Model	K-Means Clustering
Similarity Method	Cosine Similarity
Dataset	Football Players 2024–2025

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1. INTRODUCTION

Modern football generates a large amount of player performance data. Statistics such as goals, assists, minutes played, progressive actions, passing, tackling and possession-related actions can be used to describe different player profiles. A sports organization can use these profiles to support scouting and recruitment decisions.

The objective of this project is not to predict a predefined class. Instead, the system discovers natural patterns within the player statistics and then measures how close one player is to other players in the feature space. This makes the problem suitable for unsupervised machine learning and similarity analysis.

2. PROBLEM STATEMENT

A sports organization wants to identify players who have similar statistical profiles. The system should analyze football player statistics, discover groups of players with similar characteristics, compare different unsupervised learning techniques, and provide a mechanism to recommend the most statistically similar players to a selected player.

3. OBJECTIVES

- Understand and explore the football player dataset.
- Clean and preprocess the raw dataset.
- Convert multiple player records into a player-level representation.
- Select meaningful statistical performance features.
- Handle missing and infinite values.
- Scale the selected numerical features.
- Use PCA for dimensionality reduction and visualization.
- Apply multiple unsupervised clustering algorithms.
- Evaluate and compare clustering performance.
- Use cosine similarity to recommend the Top 5 similar players.
- Provide a foundation for an interactive Streamlit sports analytics application.

4. DATASET DESCRIPTION

The project uses the football players 2024–2025 dataset. The raw dataset contains player-level football statistics from the selected competition data source. The raw file used in the project contains 2,854 rows and 267 columns.

Property	Value
Raw records	2,854
Raw columns	267
Unique players	2,702
Repeated player records	152
Final performance features	39
Final player-level feature matrix	$2,702 \times 39$

Repeated player records were identified during data understanding. Instead of treating every raw record as a separate player, the final workflow aggregates the selected numerical performance features by Player using the mean. This produces one feature vector per unique player.

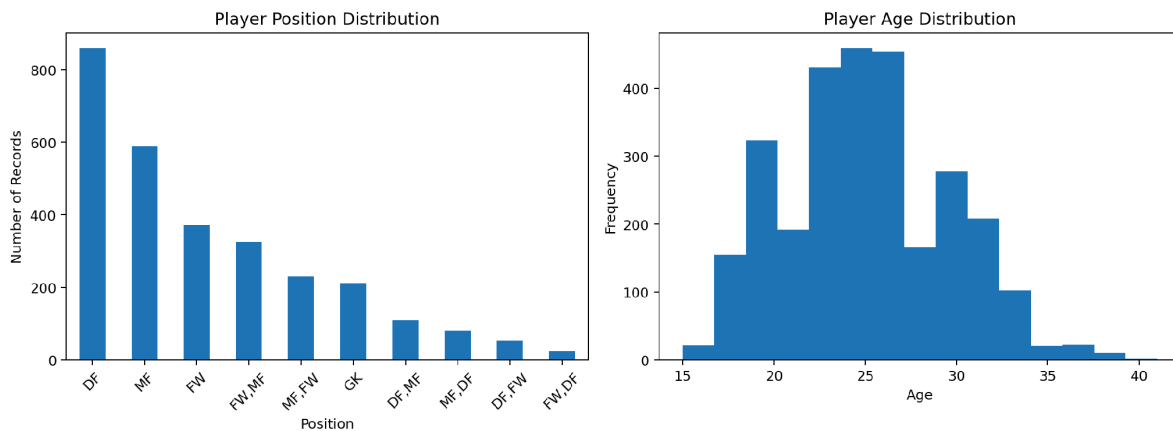


Figure 1. Player position and age distribution.

5. SYSTEM REQUIREMENTS AND TECHNOLOGIES

Technology	Purpose
Python	Core programming language
Pandas	Data loading, cleaning and transformation
NumPy	Numerical computation
Matplotlib	Static visualizations
Seaborn	Statistical visualization and heatmaps
Scikit-learn	Scaling, PCA, clustering and evaluation
Cosine Similarity	Player similarity recommendation
Streamlit	Interactive demonstration application

6. METHODOLOGY

The project follows a complete machine learning workflow. The major stages are shown below.

- Raw Dataset → Data Understanding
- Data Understanding → Exploratory Data Analysis
- EDA → Cleaning and Feature Preparation
- Feature Preparation → Player-Level Aggregation
- Player-Level Data → Missing Value Handling
- Missing Value Handling → Standard Scaling
- Scaled Features → PCA and Clustering
- Clustering → Model Evaluation and Comparison
- Scaled Features → Cosine Similarity
- Cosine Similarity → Top 5 Similar Player Recommendations

The clustering stage contains K-Means, Agglomerative Clustering, DBSCAN and Gaussian Mixture Model. The best model is selected using the evaluation results, while cosine similarity provides the actual player-to-player recommendation mechanism.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the distribution and characteristics of the dataset before applying machine learning. Position and age distributions were examined along with goals, assists and minutes played.

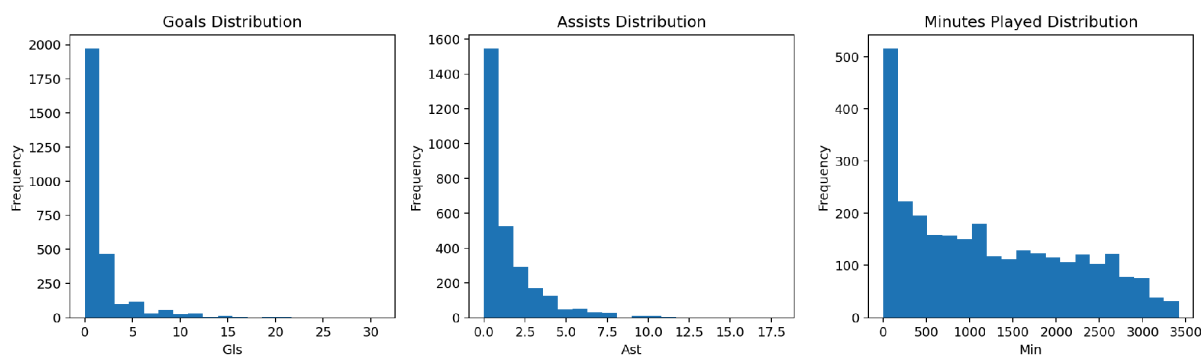


Figure 2. Distribution of goals, assists and minutes played.

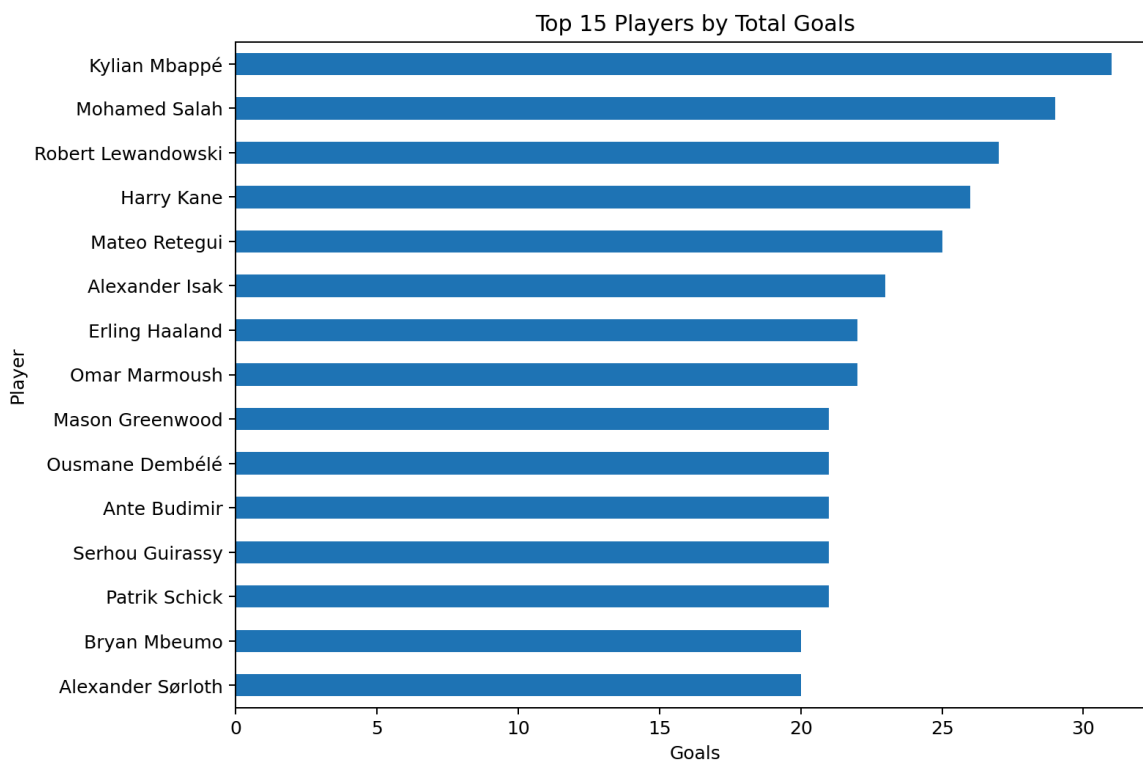


Figure 3. Top 15 players by total goals.

The analysis also examined the relationship between selected performance statistics using a correlation heatmap. This helps identify strongly related variables and provides a better understanding of the structure of the feature space.

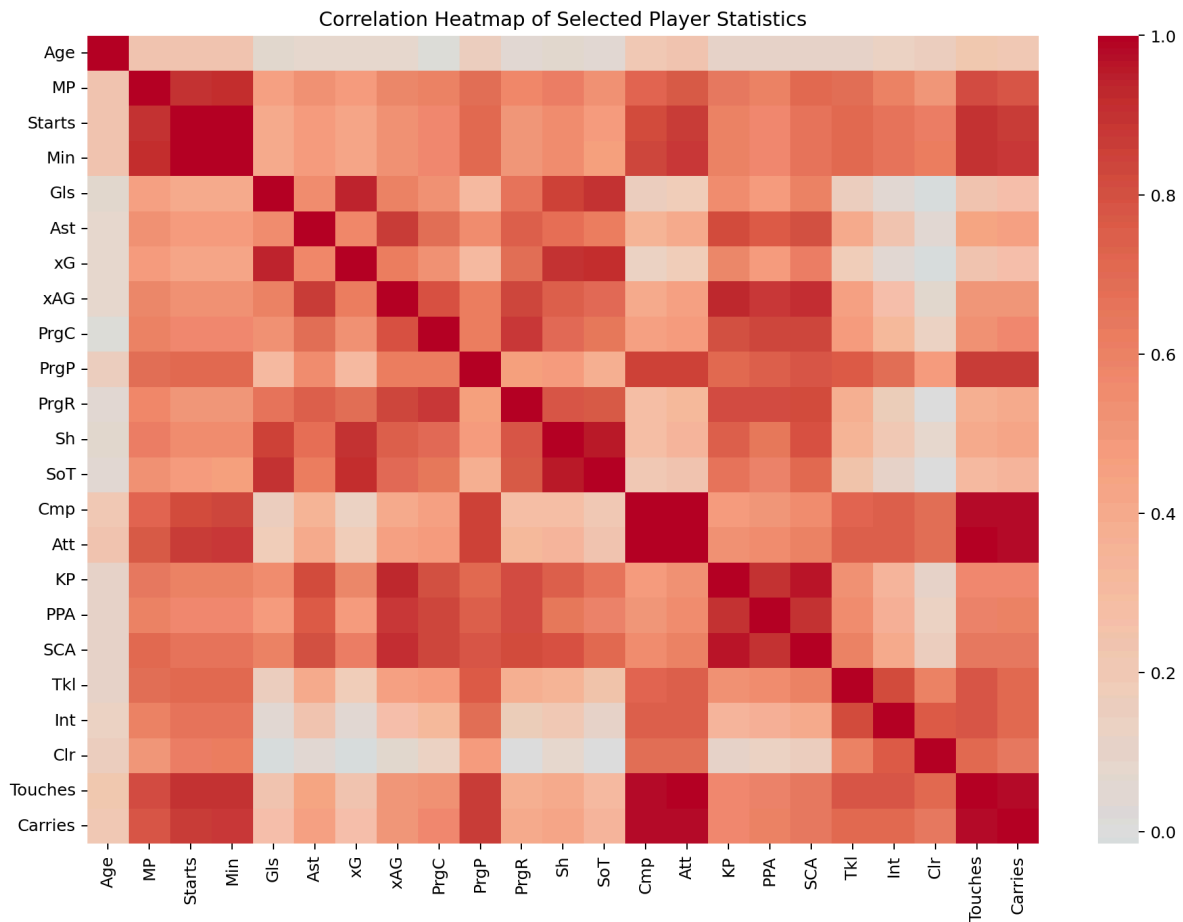


Figure 4. Correlation heatmap of selected player statistics.

8. DATA CLEANING AND PREPROCESSING

The initial cleaning stage removes exact duplicate rows, completely empty columns and infinite values. In the final dataset, no exact duplicate rows were found. Ranking columns and repeated metadata columns generated from different statistical sections were removed because they are not suitable as independent performance features.

Stage	Rows	Columns
Original dataset	2,854	267
After basic cleaning	2,854	267
After removing ranking/repeated metadata	2,854	206
Final player-level matrix	2,702	39

The final player-level feature preparation converts the selected features to numeric values, replaces infinite values with missing values, aggregates multiple records belonging to the same player, and fills remaining missing values using the median of each feature.

9. FEATURE SELECTION AND ENGINEERING

A focused set of 39 performance features was selected to represent player characteristics. The selected variables cover playing time, scoring, expected performance, progression, shooting, passing, chance creation, defending, carrying, receiving and aerial performance.

Feature Group	Examples
Playing time	MP, Starts, Min, 90s
Scoring	Gls, Ast, G+A, G-PK
Expected performance	xG, xAG, npxG+xAG, xA
Progression	PrgC, PrgP, PrgR
Shooting	Sh, SoT, SoT%
Passing & creation	Cmp, Att, Cmp%, KP, PPA, SCA, SCA90
Defending	Tkl, TklW, Int, Tkl+Int, Clr
Possession & carrying	Touches, Succ, Succ%, Carries, CPA, Rec
Aerial	Won, Won%

10. FEATURE SCALING

Feature scaling is necessary because the selected statistics are measured on different numerical ranges. For example, minutes played can have much larger values than percentages or per-90 statistics. StandardScaler transforms the features so that each feature contributes on a comparable scale.

The standardized feature matrix is used as the input for clustering and cosine similarity. This prevents features with large numeric ranges from dominating the distance or similarity calculations.

11. PCA DIMENSIONALITY REDUCTION

Principal Component Analysis (PCA) is used to reduce the 39-dimensional feature space to two principal components for visualization. PCA does not perform the clustering; it provides a compact representation that makes the overall player distribution easier to visualize.

Component	Explained Variance
PC1	49.36%
PC2	19.01%
PC1 + PC2	68.38%

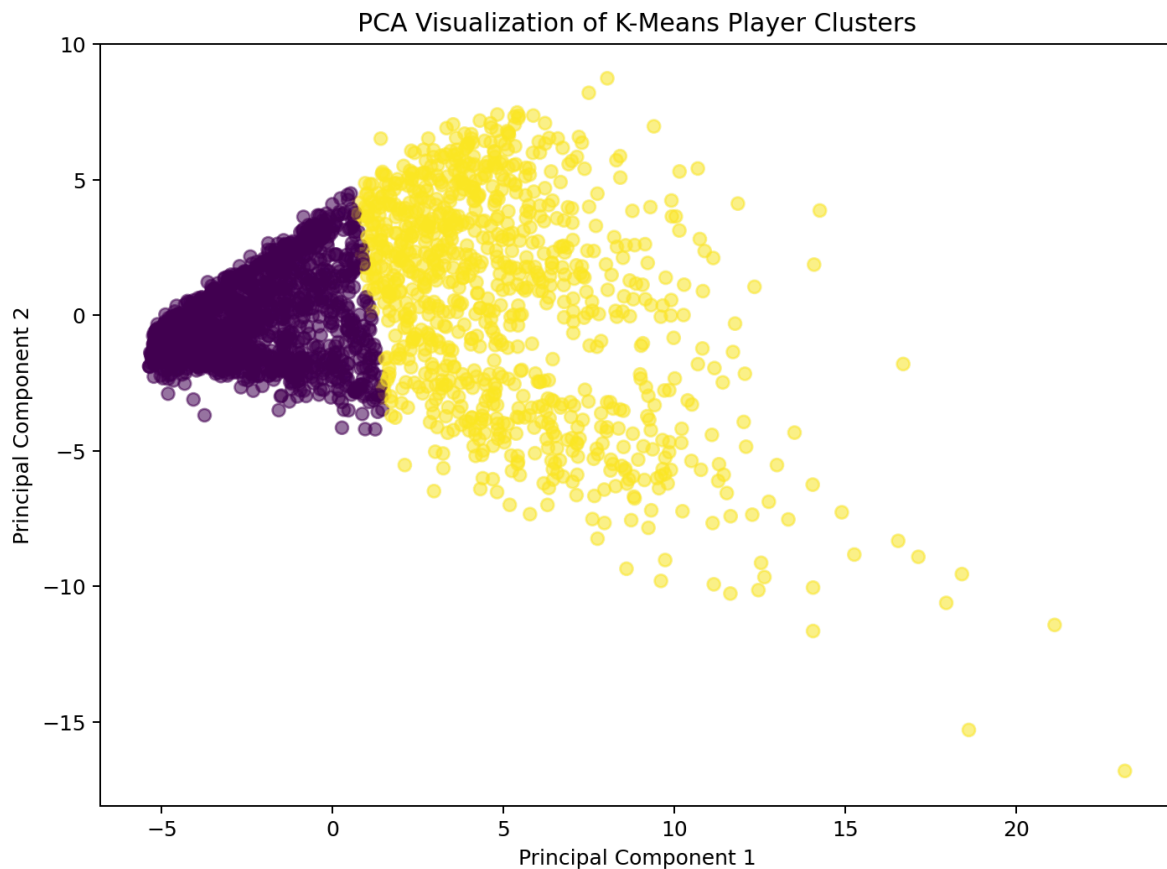


Figure 5. PCA visualization of player clusters using the K-Means labels.

12. UNSUPERVISED MACHINE LEARNING MODELS

12.1 K-Means Clustering

K-Means divides the players into a predefined number of clusters by assigning each player to the nearest cluster centroid. The project evaluates K values from 2 to 10 using inertia and silhouette score. The final workflow selected K = 2 based on the highest silhouette score.

12.2 Agglomerative Clustering

Agglomerative clustering is a hierarchical approach that starts with individual observations and progressively merges similar groups. It provides a different view of the player structure compared with centroid-based K-Means.

12.3 DBSCAN

DBSCAN is a density-based clustering technique. It can identify dense groups and noise points without requiring the number of clusters in advance. In this project, noise points are excluded when calculating the clustering metrics.

12.4 Gaussian Mixture Model

Gaussian Mixture Model represents the data using a mixture of Gaussian distributions. It provides a probabilistic clustering approach and is included to compare against the other unsupervised methods.

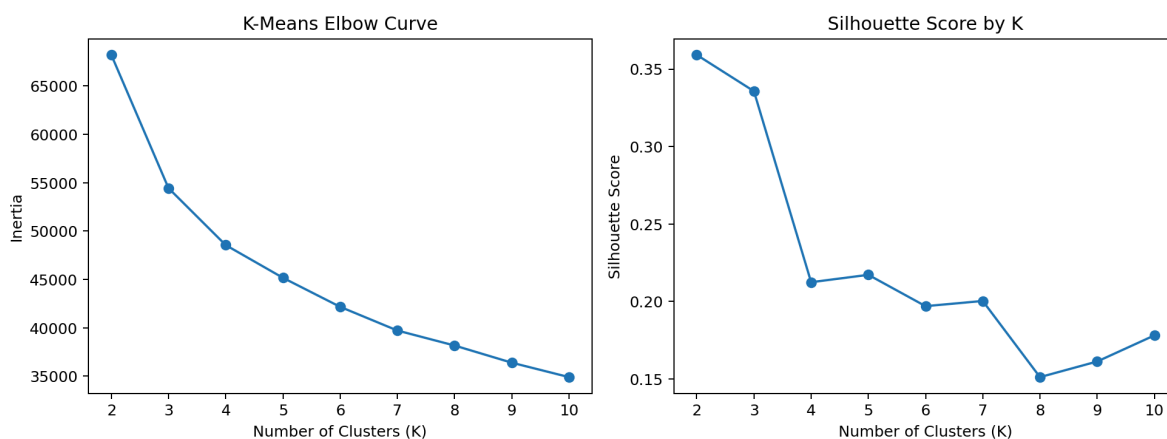


Figure 6. K-Means elbow curve and silhouette analysis.

13. MODEL EVALUATION AND COMPARISON

Three evaluation measures are used. Silhouette Score measures how well a point fits within its own cluster compared with other clusters; higher values are better. Davies-Bouldin Index measures cluster overlap; lower values are better. Calinski-Harabasz Score evaluates the ratio of between-cluster dispersion to within-cluster dispersion; higher values are better.

Model	Silhouette $\uparrow$	Davies-Bouldin $\downarrow$	Calinski-Harabasz $\uparrow$
K-Means	0.3592	1.2597	1471.25
Agglomerative	0.3572	1.3293	1290.53
DBSCAN	0.1372	1.3413	62.08
Gaussian Mixture	0.2389	1.3921	1069.55

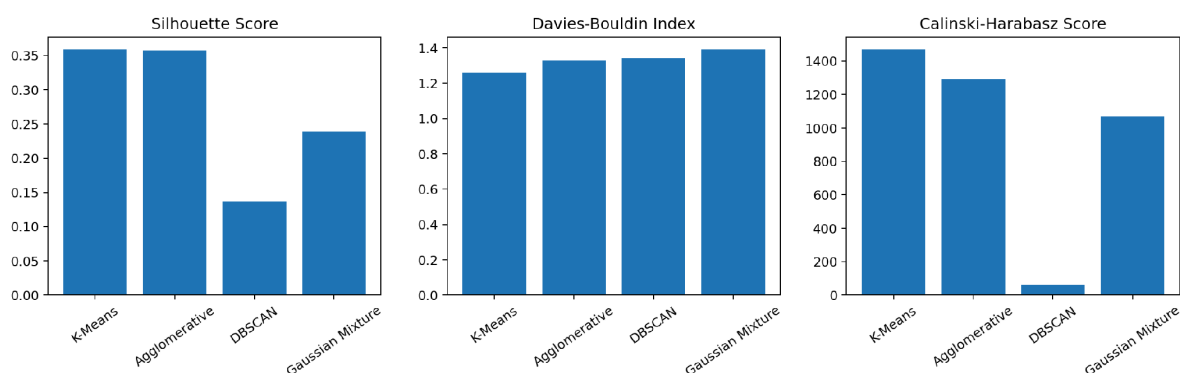


Figure 7. Comparison of the four clustering algorithms.

Based on the comparison, K-Means is selected as the best-performing clustering model for this dataset. It achieves a silhouette score of 0.3592, a Davies-Bouldin Index of 1.2597, and a Calinski-Harabasz Score of 1471.25. The other algorithms are retained in the study because comparing multiple approaches is an important part of evaluating an unsupervised machine learning solution.

14. PLAYER SIMILARITY USING COSINE SIMILARITY

Clustering identifies groups of players, but the main business requirement is to find individual players who have similar statistical profiles. For this purpose, cosine similarity is calculated between the standardized feature vectors of all unique players.

Cosine similarity measures the angle between two vectors. A value closer to 1 indicates a more similar direction in the feature space. The system removes the selected player from the result and returns the five highest similarity scores.

Rank	Player	Cosine Similarity
1	Ayanda Sishuba	0.9999
2	Sékou Doucoure	0.9997
3	Julio Soler	0.9994
4	Michael Golding	0.9993
5	Remy Rees-Dottin	0.9993

For the notebook test case, the selected player was Aaron Ciammaglichella. The recommendation function returned the Top 5 statistically similar players from the 2,702-player feature matrix.

15. RESULTS AND DISCUSSION

The final workflow successfully converts a high-dimensional football statistics dataset into a player-level machine learning representation. The preprocessing stage reduces the raw data to 2,702 unique player profiles and 39 selected performance features. Standardization ensures that different statistical scales do not dominate the clustering or similarity calculations.

The clustering comparison identifies K-Means as the strongest overall model among the four tested methods. The result is supported by the evaluation metrics rather than by a single visual inspection. Cosine similarity then adds a recommendation layer by ranking individual players according to their statistical closeness.

These outputs can support scouting and recruitment workflows by providing candidate alternatives with similar statistical profiles. However, similarity should be interpreted as statistical similarity rather than a guarantee that two players will perform identically in a tactical, financial or real-world setting.

16. STREAMLIT APPLICATION

A Streamlit-based demonstration application is planned as the presentation layer for the machine learning project. The application is designed to expose the final workflow through an interactive sports analytics dashboard.

- Dashboard showing total players, features, models and best model.
- Player Similarity page with player selection and Top 5 recommendations.
- Player Analytics page with interactive statistical visualizations.
- ML Models page explaining the four clustering algorithms.
- Model Comparison page showing evaluation metrics.
- PCA Visualization page showing player clusters in two dimensions.
- Dataset Explorer for filtering and inspecting player-level data.
- About Project page documenting the case study and methodology.

The Streamlit application should reproduce the same preprocessing and similarity logic used in the final notebook. It should not use fabricated recommendations or hard-coded similarity scores.

17. LIMITATIONS

- Statistical similarity does not capture tactical role, team system, leadership, injury history, contract value or other non-statistical factors.
- The project depends on the quality and completeness of the available football statistics.
- Different feature selections or aggregation strategies can change the clustering structure.
- DBSCAN performance is sensitive to parameter choices such as `eps` and `min_samples`.
- Cosine similarity is based on the selected numerical feature space and should be treated as a recommendation aid rather than a final scouting decision.

18. FUTURE SCOPE

- Deploy the Streamlit application online for interactive scouting demonstrations.
- Add player-position-specific similarity models so that defenders are compared primarily with defenders, midfielders with midfielders, and so on.
- Include additional contextual features such as league strength, tactical role and age progression.
- Experiment with advanced dimensionality reduction and embedding methods.
- Add filters for age, position, league, squad and playing style.
- Build a scouting shortlist export feature.
- Add historical season comparison to identify player development trends.

19. CONCLUSION

This project demonstrates a complete unsupervised machine learning workflow for football Player Similarity Analysis. The raw dataset of 2,854 records was transformed into 2,702 unique player profiles using player-level aggregation. Thirty-nine performance features were selected, missing values were handled, and the data was standardized before machine learning.

Four clustering approaches were compared: K-Means, Agglomerative Clustering, DBSCAN and Gaussian Mixture Model. Based on the evaluation metrics, K-Means provided the strongest overall clustering performance. PCA was used to visualize the player distribution, while cosine similarity was used to produce Top 5 similar-player recommendations.

The resulting system provides a practical foundation for sports scouting and recruitment support. It demonstrates how machine learning can convert raw player statistics into meaningful groups and similarity-based recommendations while also providing an interactive path through a Streamlit application.

20. REFERENCES

- Scikit-learn Documentation — Clustering, PCA, preprocessing and model evaluation.
- Pandas Documentation — DataFrame manipulation and data preprocessing.
- Matplotlib Documentation — Data visualization.
- Seaborn Documentation — Statistical data visualization.
- Streamlit Documentation — Building interactive Python data applications.
- Football Players 2024–2025 dataset used for this project.